Short Questions 7
03/08/2026

Your answers should be formal with mathematical formulas and algorithms, as necessary. Upload a PDF file. If you want to add a drawing, you can include a nicely hand-drawn figure or a figure drawn with a computer tool. Use single spaced typing. Preferably, keep your answers to a page.

Type your responses. Starting this set of questions, I will take off points if you don't make any attempt to submit typed answers.

1. (2 points) What are arguments for going beyond one step in a temporal difference (TD) learning by an agent that learns by interacting with an environment?
2. (3 points) Write the formula for the return G_t for an n-step SARSA-like TD algorithm for the two cases: i) when it is used to estimate $v(s)$, and ii) when it is used to estimate $q(s, a)$.
3. (5 points) Traditional reinforcement learning algorithms are table-based that store $v(s)$ and/or $q(s, a)$ values and update them. List some problems with such table-based reinforcement learning algorithms. How can attempts be made to overcome some of these problems? How does machine learning, in particular, deep learning, may be of help?